

WASP-12b

R-0359

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_231125 · created 2026-07-18T12:38:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460274.9432 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.045
Transit depth (Rp/Rs)^2	2.19 +/- 1.35 [%]
Orbital Inclination (inc)	85.27 +/- 2.92
Airmass coefficient 1 (a1)	3.29 +/- 1.9
Airmass coefficient 2 (a2)	-1.01 +/- 0.95
Scatter in the residuals of the lightcurve fit is	65.96 %
Best Comparison Star	None
Optimal Aperture	2.53
Optimal Annulus	4.0
Transit Duration (day)	0.1301 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.045 vs 0.1178 (deviation 0.54σ)
Duration fit vs published	0.1301 d vs 0.125 d (deviation 0.59σ)
Mid-time O–C	1568.0 min
Depth SNR	2.7
Residual scatter	65.96 %

Light curve

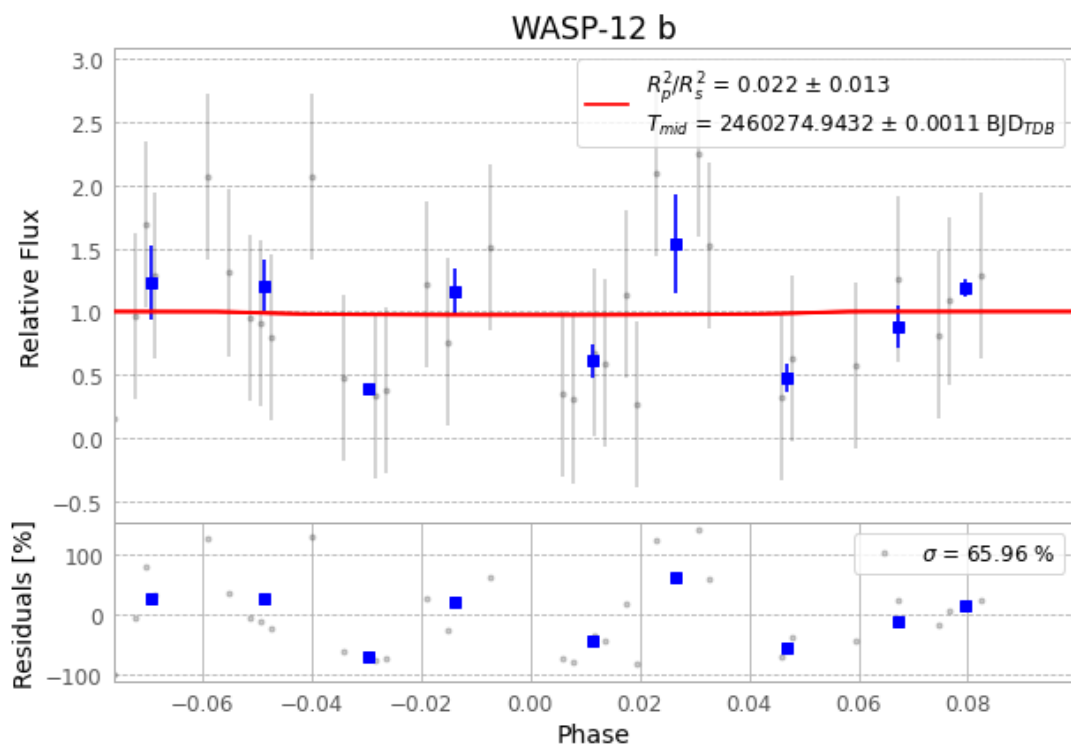


Figure 1 — FinalLightCurve_WASP-12 b_2023-11-26.png (SHA-256 bb13a87cceb64a75...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [392, 339], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1f09b10a481ed016...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9e597cf

Data provenance

93 FITS frames (61 MB), WASP-12231126083015.FITS → WASP-12231126130615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231125.log (ad37843487d6...), FinalLightCurve_WASP-12 b_2023-11-26.png (bb13a87cceb6...), FinalLightCurve_WASP-12 b_2023-11-26.csv (02f4191ac159...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 12:38Z.